Writeup

Open the PCAP File

First, open the provided file using Wireshark

The capture contains around 1,600 packets.

There is a lot of normal traffic, so we should not inspect every packet one by one.

We start with basic network analysis.

Check DNS Traffic

Use this Wireshark display filter:

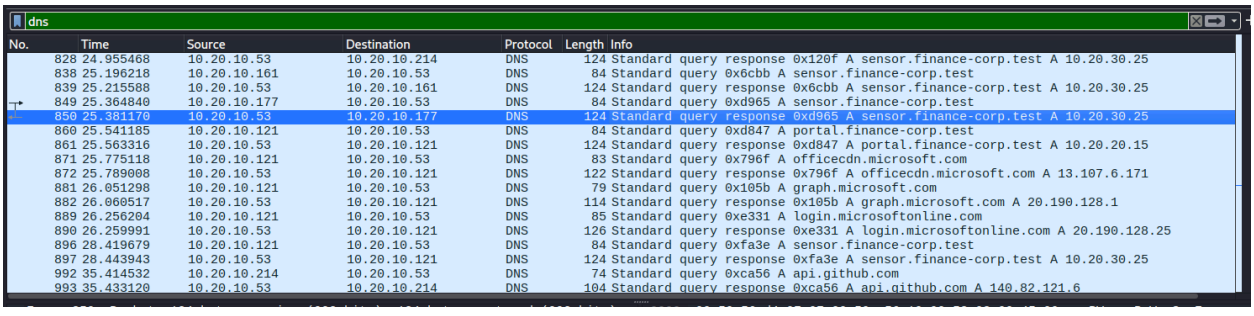
```
dns
```

DNS traffic shows the domain names requested by the different workstations.

We can see many normal domains such as Microsoft, Google, and GitHub.

However, one internal domain looks interesting:

```
sensor.finance-corp.test
```



The word `sensor` suggests that this could be related to an endpoint security or EDR system.

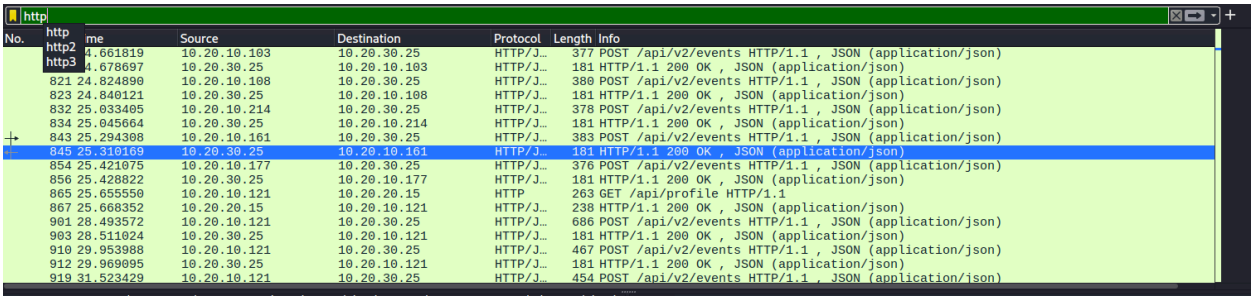
Because the challenge mentions a security alert, this is a useful place to investigate.

Check HTTP Traffic

Next, change the Wireshark filter to:

```
http
```

We can see multiple HTTP requests being sent to the internal sensor server.



Some requests use API paths such as:

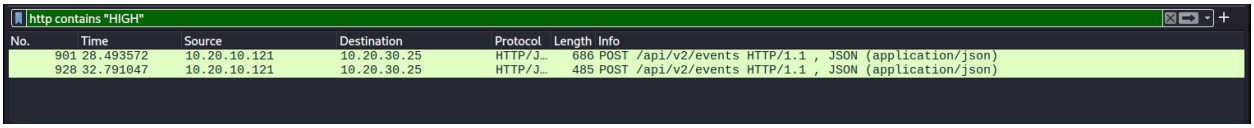
```
/api/v2/events
/api/v2/heartbeat
/api/v2/artifacts/
```

This looks like endpoint security telemetry.

Instead of checking every event manually, we can search for high-severity alerts.

Use:

```
http contains "HIGH"
```



This reduces the amount of traffic and shows only the important security events.

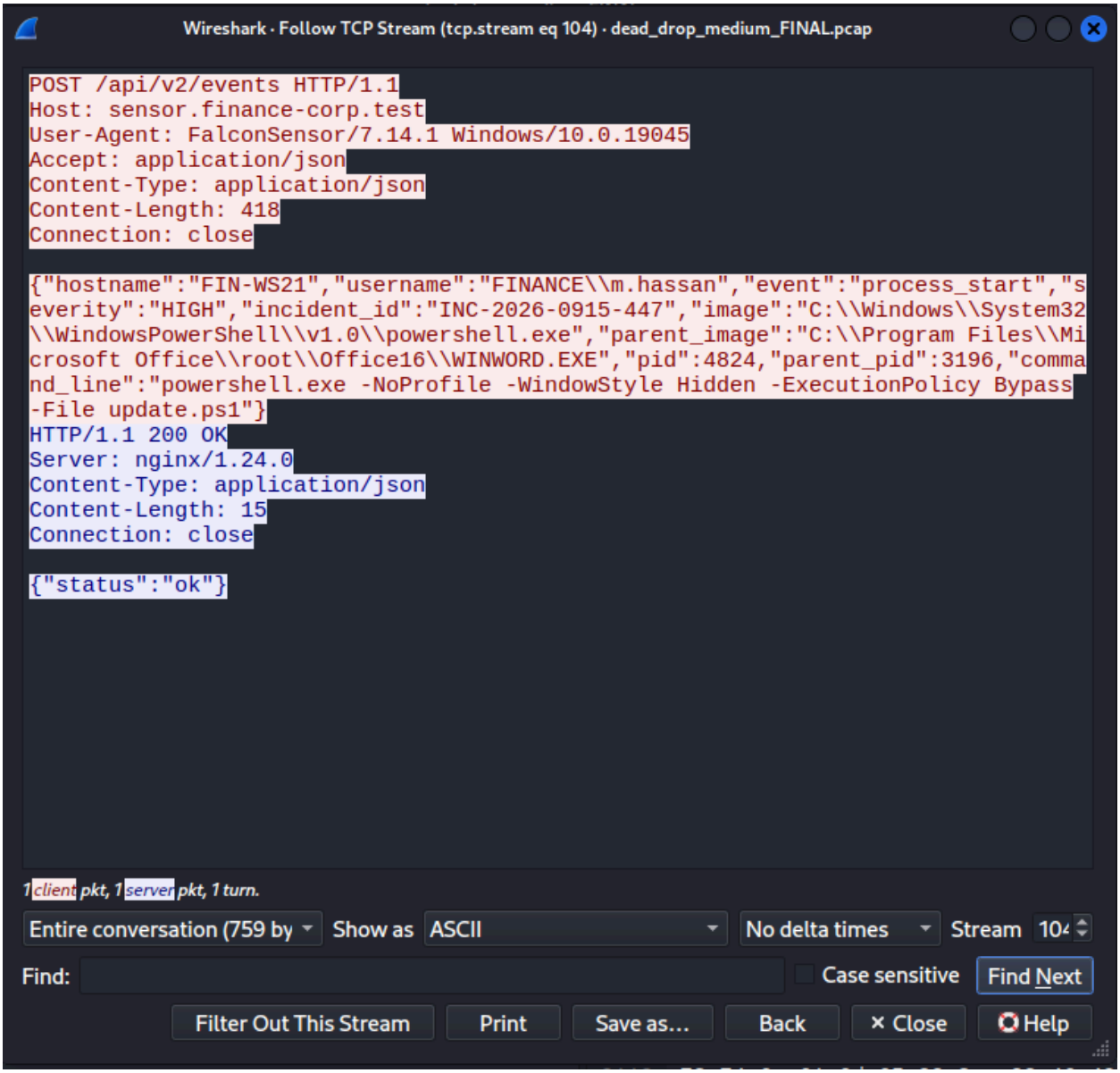
Identify the Compromised Workstation

Select one of the packets found by the previous filter.

Right-click the packet and choose:

```
Follow → TCP Stream
```

Inside the TCP stream, we can see JSON data similar to:




```
{
  "hostname":"FIN-WS21",
  "username":"FINANCE\\m.hassan",
  "event":"process_start",
  "severity":"HIGH",
  "incident_id":"INC-2026-0915-447",
  "image":"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",
  "parent_image":"C:\\Program Files\\Microsoft Office\\root\\Office16\\WINWORD.EXE",
  "pid":4824,
  "parent_pid":3196,
  "command_line":"powershell.exe -NoProfile -WindowStyle Hidden -ExecutionPolicy Bypass -
```

```
File update.ps1"
}
```

From this event, we learn several important things:

```
Hostname: FIN-WS21
User: FINANCE\m.hassan
Severity: HIGH
Incident ID: INC-2026-0915-447
```

We also see a suspicious process chain:

```
WINWORD.EXE
     ↓
powershell.exe
```

Microsoft Word starting PowerShell with hidden execution is suspicious.

The most important information here is the incident ID:

```
INC-2026-0915-447
```

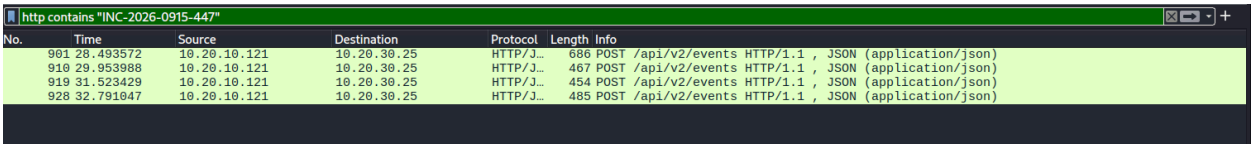
We can now use this ID to find all events related to the same incident.

Follow the Incident ID

Close the TCP stream window.

Use this Wireshark filter:

```
http contains "INC-2026-0915-447"
```

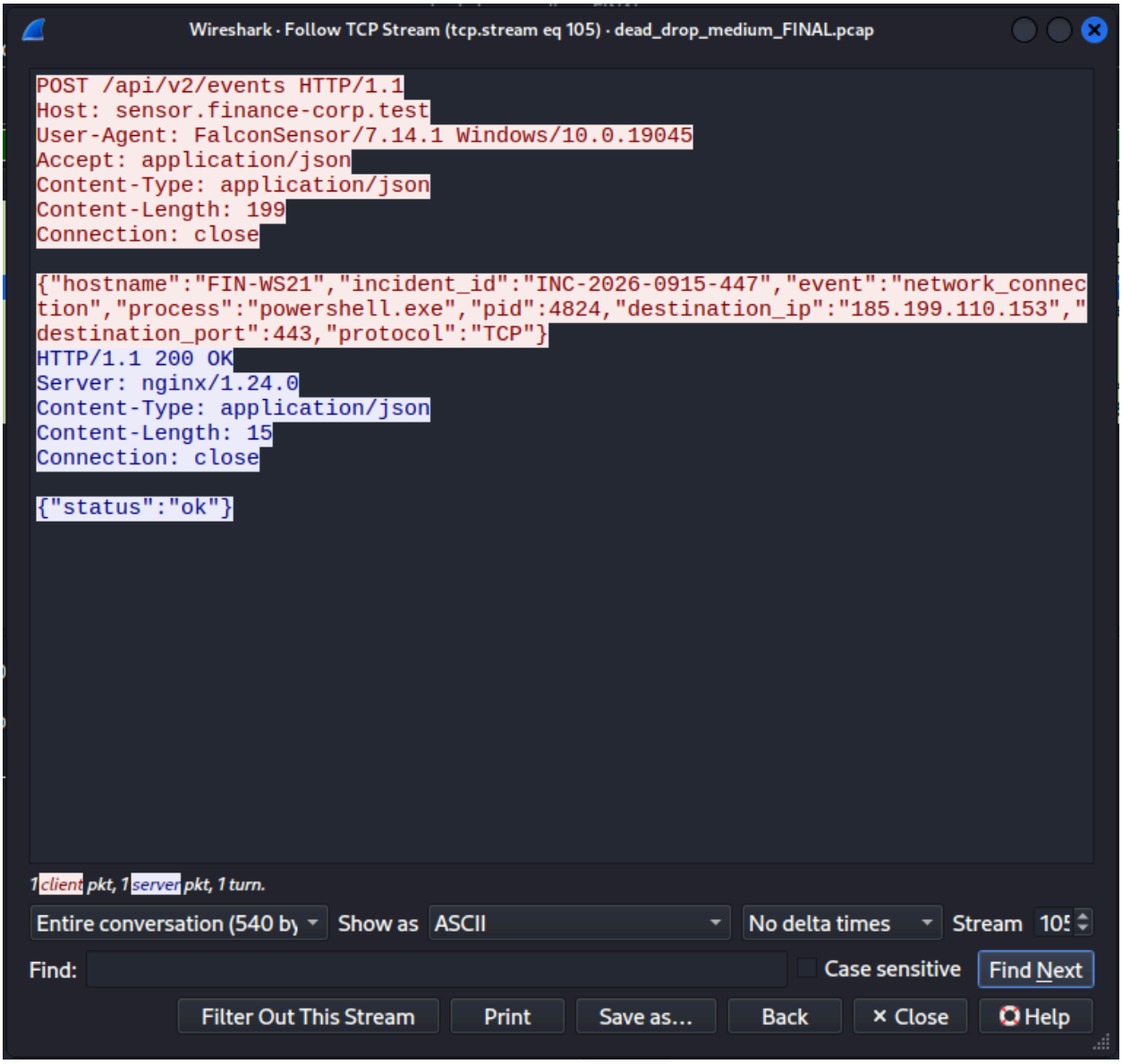


This shows multiple events related to the compromised workstation.

We can inspect the events one by one using:

```
Right-click → Follow → TCP Stream
```

One event shows a network connection:




```
{
  "hostname":"FIN-WS21",
```

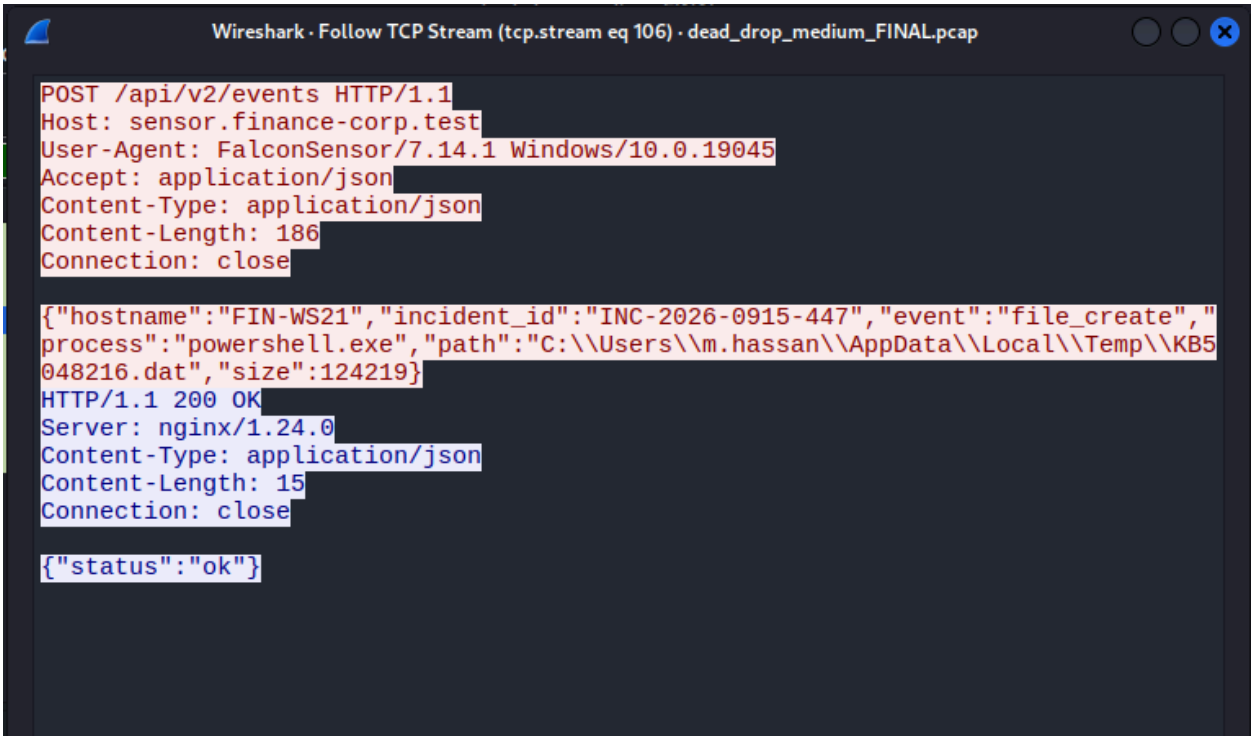
```
  "incident_id":"INC-2026-0915-447",
  "event":"network_connection",
  "process":"powershell.exe",
  "pid":4824,
  "destination_ip":"185.199.110.153",
  "destination_port":443,
  "protocol":"TCP"
}
```


This tells us that the suspicious PowerShell process made an outbound connection.

This supports the idea that the PowerShell process was part of the malicious activity.

Find the Suspicious File

Another event from the same incident shows a file creation:




```
{
  "hostname":"FIN-WS21",
  "incident_id":"INC-2026-0915-447",
  "event":"file_create",
```

```
  "process":"powershell.exe",
  "path":"C:\\Users\\m.hassan\\AppData\\Local\\Temp\\KB5048216.dat",
  "size":124219
}
```


Now we know the suspicious file is:

```
KB5048216.dat
```

The full path is:

```
C:\Users\m.hassan\AppData\Local\Temp\KB5048216.dat
```

Even though the extension is `.dat`, the file is connected to the suspicious PowerShell activity.

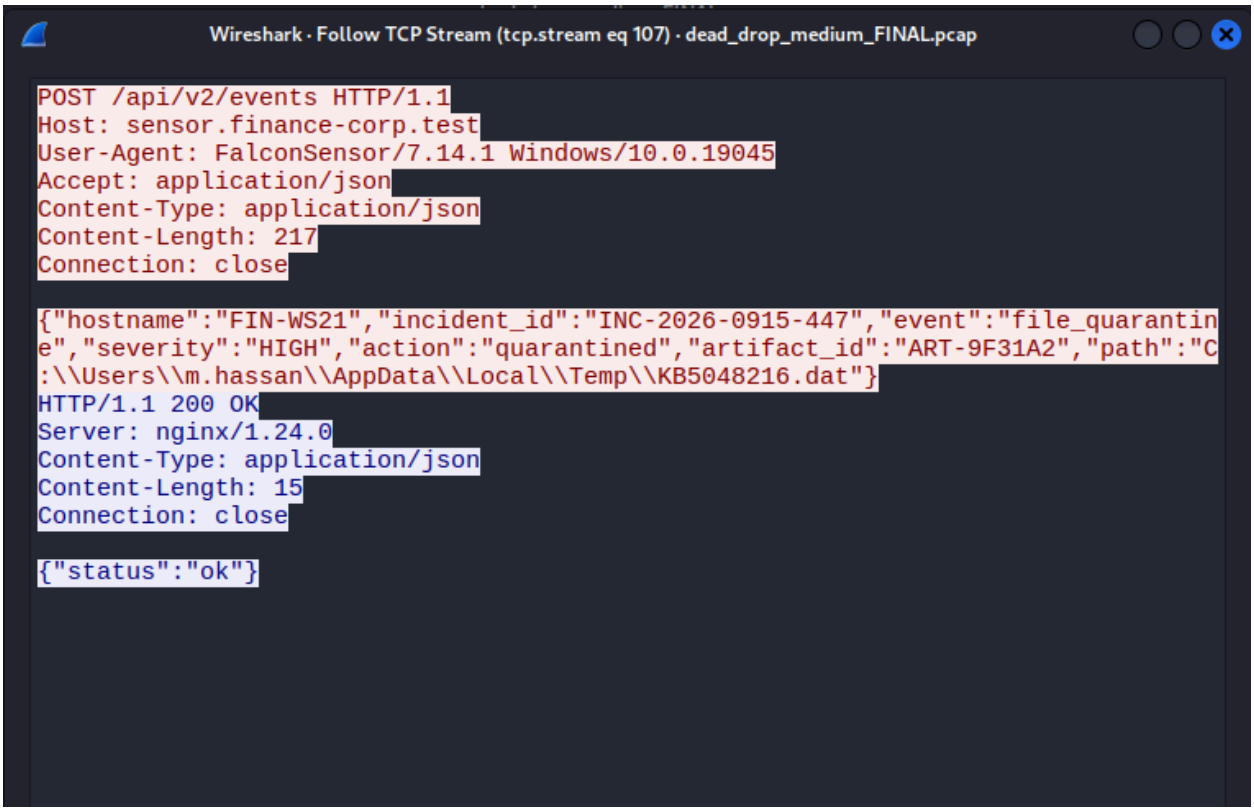
This makes it an important artifact.

Find the Quarantine Event

Continue checking the events related to:

```
INC-2026-0915-447
```

We find another event:




```
{
  "hostname":"FIN-WS21",
  "incident_id":"INC-2026-0915-447",
  "event":"file_quarantine",
  "severity":"HIGH",
  "action":"quarantined",
  "artifact_id":"ART-9F31A2",
  "path":"C:\\Users\\m.hassan\\AppData\\Local\\Temp\\KB5048216.dat"
}
```

This confirms that the suspicious file was quarantined by the security system.

The important new information is:

```
Artifact ID: ART-9F31A2
```

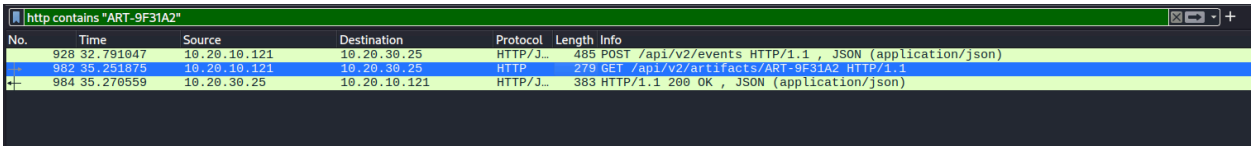
Since the file itself is no longer available, we can use this artifact ID to find information about it.

Search for the Artifact ID

Use this Wireshark filter:

```
http contains "ART-9F31A2"
```

We can see a request similar to:




```
GET /api/v2/artifacts/ART-9F31A2
```

Follow the TCP stream:

```
Right-click → Follow → TCP Stream
```

The HTTP response contains information about the quarantined file:



```
{
  "artifact_id":"ART-9F31A2",
  "filename":"KB5048216.dat",
  "size":124219,
  "algorithm":"SHA256",
  "digest":"f34a962044d095bb136b8523d3f51fd665a39f2090aa9b5ba1aba8d9df0642f5",
  "classification":"unknown",
  "action":"quarantined"
}
```


We now have the SHA256 hash:

```
f34a962044d095bb136b8523d3f51fd665a39f2090aa9b5ba1aba8d9df0642f5
```

Why This Hash Is the Correct One

The PCAP contains other hashes as background noise.

Because of that, we should not simply search every SHA256 value we find.

We know this is the correct hash because we followed the complete incident chain:

```
FIN-WS21
   ↓
INC-2026-0915-447
   ↓
powershell.exe
   ↓
KB5048216.dat
   ↓
file_quarantine
   ↓
ART-9F31A2
   ↓
SHA256
```

This makes the investigation evidence-based instead of guessing.

Investigate the SHA256 on VirusTotal

Copy the SHA256:

```
f34a962044d095bb136b8523d3f51fd665a39f2090aa9b5ba1aba8d9df0642f5
```

Search the hash on VirusTotal.

After opening the file report, go to the:

```
Behavior
```

section.

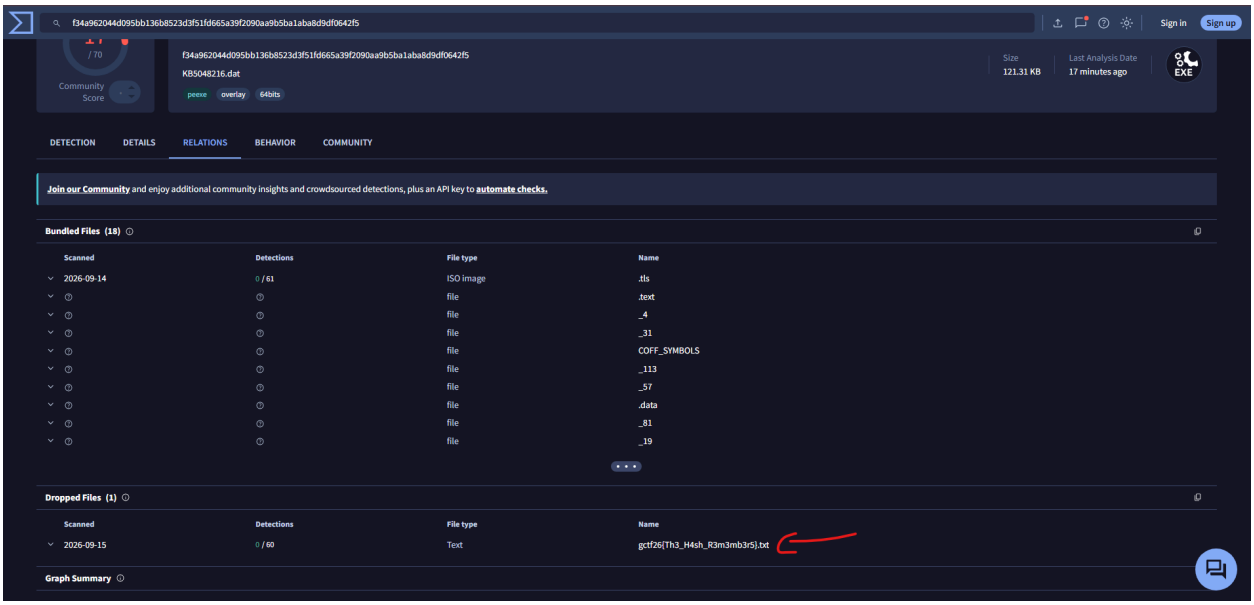
Look for file activity such as:

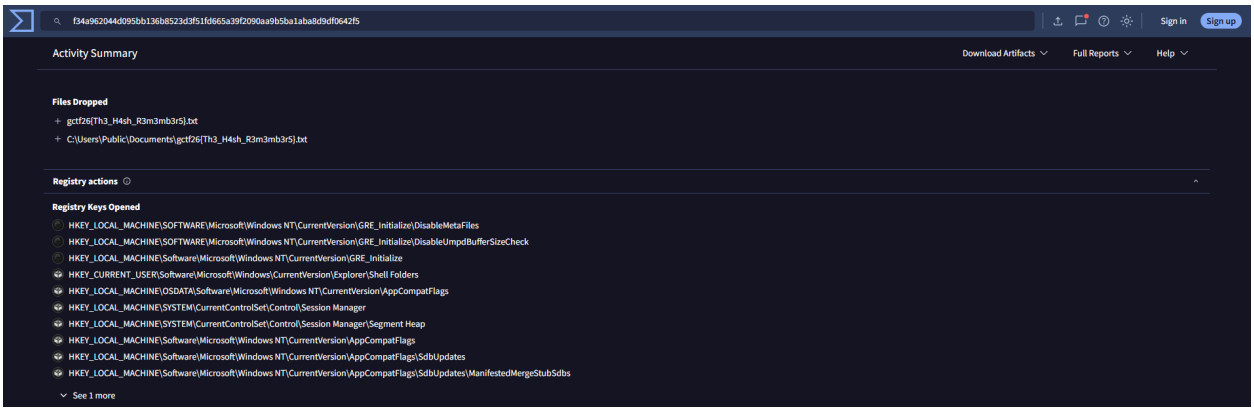
```
Files Written
```

The sandbox behavior shows the following file path:

```
C:\Users\Public\Documents\gctf26{Th3_H4sh_R3m3mb3r5}.txt
```

The filename contains the flag directly.





Flag

```
gctf26{Th3_H4sh_R3m3mb3r5}
```